

RESEARCH ARTICLE

Joint species distribution modelling for restoration of American chestnut (*Castanea dentata*) in the United States under current and future climate conditions

Matthew E. Fertakos¹  | Bethany A. Bradley²  | Jeffery W. Doser³  |
Malcolm S. Itter² 

¹Graduate Program in Organismic and Evolutionary Biology, University of Massachusetts, Amherst, Massachusetts, USA

²Department of Environmental Conservation, University of Massachusetts, Amherst, Massachusetts, USA

³Department of Forestry and Environmental Resources, North Carolina State University, Raleigh, North Carolina, USA

Correspondence

Matthew E. Fertakos

Email: mfertakos@umass.edu

Funding information

National Science Foundation Graduate Research Fellowship Program, Grant/Award Number: 1938059; Northeast Climate Adaptation Science Center, University of Massachusetts Amherst, Grant/Award Number: G23AC00519

Handling Editor: Juan C. Álvarez-Yépiz

Abstract

1. American chestnut (*Castanea dentata*) was once a dominant tree species spanning the eastern temperate forests of the United States until the introduction of chestnut blight in the early twentieth century resulted in its functional extinction. Recent advances in the development of blight-resistant American chestnut could support widespread restoration, but optimal locations for restoration plantings remain uncertain.
2. Previous models of American chestnut created to address this question rely on post-blight occurrence data and pseudo-absences which likely underrepresent its distribution. They also omit abundance information, which may be critical for identifying successful restoration sites. Additionally, climate change is shifting the locations of suitable habitat for eastern temperate trees like American chestnut, complicating long-term restoration planning.
3. We use a Bayesian joint species distribution modelling (JSDM) framework to jointly model current and future suitability for 11 chestnut-associated tree species within a broader assemblage of 39 tree species commonly found in eastern temperate forests.
4. We identified a poleward shift in habitat for the forest communities American chestnut historically occurred with. Especially suitable areas were in the northeastern United States and northern Michigan, as well as possible refugia in higher elevation areas of the Appalachian Mountains. These areas overlap with 2988 existing conservation properties most suitable for restoration.
5. *Synthesis and applications.* Our joint species distribution modelling approach highlights its value in guiding the adaptive restoration of functionally extinct species in forests shaped by past and ongoing disturbance. We predicted a northward shift in species associated with American chestnut and identified lands already set aside for conservation where conditions are likely to be suitable for its long-term survival under climate change. These results provide actionable guidance

for expanding experimental assisted migration and restoration plantings of blight-resistant American chestnut throughout and beyond its historical range.

KEYWORDS

American chestnut, Bayesian hierarchical modelling, conservation plantings, eastern temperate forests, forest adaptation, restoration

1 | INTRODUCTION

The American chestnut (*Castanea dentata*) was a dominant tree species spanning the eastern temperate forest ecoregion of the United States until it was nearly eradicated by blight. Where they were most abundant, American chestnut is estimated to have composed up to 50% of the forest (Russell, 1987) and could grow to be 12 feet wide and over 100 feet tall. Often described as ‘the perfect tree’, American chestnuts provided an abundant and nutritious food source for people and livestock, and its rot and insect resistant wood served as strong timber for building (Davis, 2005). Ecologically, it filled a unique role in the function of eastern temperate forests, rapidly sequestering carbon and contributing nutrients to the soil (Ellison et al., 2005), as well as producing prolific masts with easily accessible nuts used by species across trophic levels (Dalglish & Swihart, 2012). Today, less than 10% of the original population remains across its range (Dalglish et al., 2016) after the introduction of chestnut blight (*Cryphonectria parasitica*) in the late 19th century, leaving those that persist as small non-reproductive root sprouts in a perpetual cycle of resprouting, infection, and stem dieback (Anagnostakis, 1995).

Replanting of blight-resistant American chestnut could restore its lost ecological and sociological functions. Blight-resistant hybrids (Jacobs, 2007) and genetically modified trees (Newhouse & Powell, 2021) have performed well in a range of experimental reintroductions (Clark et al., 2022). However, restoring American chestnut regionally and at a landscape scale requires that conservation practitioners identify a more comprehensive number of sites well-suited for reintroduction within existing conservation lands. The designation ‘well-suited’ must include consideration of abiotic conditions, such as the need for species ranges to shift in response to climate change (Sittaro et al., 2017; Zhu et al., 2012), as well as biotic conditions, such as the forest community composition historically associated with American chestnut populations.

Many spatial models have already identified climatically suitable potential habitat for American chestnut range wide (e.g. Adeyemo & Granger, 2023; Barnes & Delborne, 2019; Noah et al., 2021), and at smaller scales (Henderson et al., 2022, 2023; Klaus & Klaus, 2025; Tulowiecki, 2020). For example, Adeyemo and Granger (2023) use a single species distribution model (SDM) with 1806 modern occurrences in combination with temperature, precipitation, soil, and elevation to predict the presence and absence of American chestnut across its native range under current and future conditions. However, this type of modelling approach may be limited by a reliance only on

post-blight distributional data, introducing spatial biases. Current American chestnut occurrences undercount and may misrepresent its original broader and more abundant distribution (Burke, 2012). Many previous SDM approaches also rely on pseudo-absence data instead of true absences, which could be particularly problematic when representative distributional data are lacking creating models with worse fit (Wisiz & Guisan, 2009). Additionally, given the former dominance of American chestnut, information about potential abundance of the species could be critical for identifying sites for restoration with a higher likelihood of success—abundance information is not included in previous SDMs (Adeyemo & Granger, 2023; Barnes & Delborne, 2019; Noah et al., 2021).

Joint species distribution models (JSDMs) address many of these gaps yet remain underutilized in restoration ecology. As opposed to traditional SDMs, which predict distributions based on fixed environmental effects and a residual site-level random effect, JSDMs additionally account for statistical dependence among species in their responses to included environmental variables and in site-level random effects (Clark et al., 2014; Pollock et al., 2014; Warton et al., 2015). Joint modelling approaches generally result in improved prediction of local communities because they account for the fact that species do not occur independently of one another at a site (Poggiato et al., 2021). While species assemblages of North American forests are not fixed (Williams et al., 2001) and may respond differently to a changing climate (Gougherty et al., 2024; Hobbs et al., 2018), co-occurrence emerges from shared environmental filtering and trait similarity, allowing assemblages to act as indirect indicators to estimate suitable habitat conditions (Ovaskainen et al., 2017). For example, Zhang et al. (2020) compared single and joint modelling of 31 prevalent and 6 rare marine species and found community data in the joint models improved the prediction of rare species occurrence. This methodology is similar to a successful conservation approach at regional and landscape scales using ‘surrogate’ species, in which a smaller group of well-known species is used as a proxy for those harder to survey (Tälle et al., 2023). Such an approach is especially useful for a species like American chestnut, which has limited extant distributional data due to near extirpation after the introduction of blight. We can take advantage of the joint abundance of species associated with American chestnut and available robust sampling data of the forests across its range to make estimates of its potential modern abundance. Areas predicted to have a high abundance of chestnut-associated species would then serve as a starting point for reintroduction efforts.

In this study, we ask which sites and associated environmental conditions across eastern North America are most likely to support forest

communities in which American chestnut could thrive under current and future climatic conditions. Using a two-stage Bayesian joint species distribution model, we examine where species historically associated with American chestnut are predicted to occur and, conditional on their presence, where overstory tree abundance is expected to be greatest. We further ask how projected climate warming scenarios alter these patterns and what implications these changes have for identifying optimal locations for chestnut introduction. Finally, we evaluate the extent to which high-priority restoration areas overlap with existing conservation lands, thereby identifying opportunities and gaps for targeted silvicultural experiments and management of existing forests.

2 | METHODS

2.1 | Data assembly

2.1.1 | FIA data

USDA Forest Service Forest Inventory and Analysis (FIA) data were downloaded using the 'rFIA' package (Stanke et al., 2020) in R v. 4.4.0 (R Core Team, 2024) for 29 states intersecting and surrounding the pre-blight range of American chestnut (Figure 1b, Little, 1971). These data were then further filtered to only include plots within FIA designated forest groups containing species historically associated with American chestnut. Plots were then filtered to the pre-blight range of American chestnut (Little, 1971) with a 10km buffer to better capture the species assemblages where chestnut once occurred. Rare species which lack sufficient data for effective modelling were excluded by keeping only those present in at least 5% of the plots across the dataset. The most recent inventory measurement was used for plots measured in more

than 1 year. This resulted in a total of $n=25,416$ plots, and 39 species for use in subsequent modelling.

2.1.2 | Selection of chestnut-associated species

Species historically associated with American chestnut made up a subset of species included in the model and were determined based on a review of natural history literature from the region (Braun, 1950; Whitney, 1994), past studies (Cogbill et al., 2002; Hanberry & Nowacki, 2016; Korstian, 1927; Paillet, 2003; Russell, 1987) and proceedings (Steiner & Carlson, 2006) investigating the communities within which American chestnut historically occurred. From the literature, we developed a candidate list of contemporary associated tree species. American chestnut may have had different associations through paleo history, so we also investigated associations in the fossil pollen record. Fossil pollen data since the last glacial maximum (22,000 BP) were downloaded from the Neotoma Paleoecology Database (Williams et al., 2018) using the 'neotoma2' R package (Dominguez & Goring, 2023). To evaluate their associations over time, we calculated the proportion of plot-year combinations within our study area that contained *Castanea* pollen and pollen from each genus of our potential chestnut-associated species identified from the literature (*Acer*, *Betula*, *Carya*, *Fagus*, *Liriodendron*, and *Quercus*). Genera which had little pollen association with *Castanea* were removed (e.g. '*Liriodendron*'). Therefore, we selected chestnut-associated species with the greatest certainty as those species that were mentioned >2 times to co-occur with American chestnut in the literature and also showed consistent association at the genus level in the pollen record. The final set of chestnut-associated species includes *Acer rubrum*, *Betula lenta*, *Carya cordiformis*, *Carya glabra*, *Carya ovata*, *Fagus grandifolia*, *Quercus alba*, *Quercus coccinea*, *Quercus prinus*, *Quercus rubra* and *Quercus velutina*.

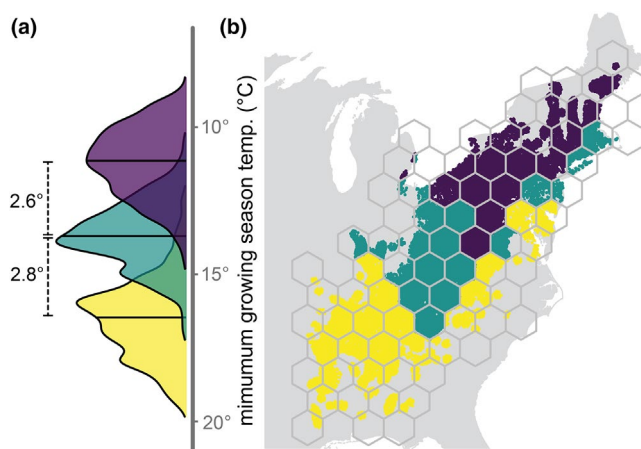


FIGURE 1 (a) Density plot of mean minimum temperature over the growing season (May–September) for each FIA plot. Medians are indicated by black lines. (b) FIA plots filtered to the range of American chestnut (Little, 1971) used to train the model coloured by fold identity. Hexagon blocks used to assign FIA plots into folds represent the range of spatial autocorrelation (155 km) are overlaid in grey.

2.2 | Environmental predictor variables

A variety of environmental variables commonly used to model tree species distributions were considered for inclusion (Iverson et al., 2019). These included growing season (May–September): minimum/maximum temperature, total precipitation, potential and actual evapotranspiration, climate water deficit, vapour pressure deficit, and soil moisture downloaded at a 4 km resolution from TerraClimate for current conditions (Abatzoglou et al., 2018), as well as for future mean increase in global temperatures of +2°C and +4°C above preindustrial levels (Qin et al., 2020). We also considered several soil variables at a 100 m resolution including percent sand content, pH, and organic soil content (Ramcharan et al., 2018) along with elevation, slope, and aspect at a 305 m resolution (Terrain Tiles, 2026). Stage 2 models also included measures of total basal area per hectare (BAH) and quadratic mean diameter (diameter of tree of average basal area; QMD) derived from the FIA data at each site as a measure of forest density and size to control for successional niche dynamics potentially

affecting the relative abundance of chestnut-associated species (see [Appendix S1](#) for more detail).

Exploratory modelling for a subset of species associated with American chestnut was used to identify a subset of environmental variables with low among-variable correlation and high explanatory power for species presence/absence. Based on this exploratory modelling, we focused on a reduced set of environmental variables representing climate (minimum temperature, precipitation), physiographic (pH, sand content, organic soil content, slope, elevation) and biophysical (climatic water deficit) conditions.

2.3 | Modelling approach

We applied a two-stage Bayesian joint species distribution modelling (JSDM) approach to model the basal area of each of 39 observed species across FIA plots in the study region. The modelling approach was adapted from a recently developed model to predict site-specific, species-level live aboveground biomass applying FIA data (Doser et al., 2025). This approach applies a generalized linear mixed model with the mean probability of presence or abundance estimated as a function of the environmental predictors and random effects that account for residual species dependence. The hierarchical structure of the model also accounts for dependencies among species-level responses to included environmental predictors. In the first stage (Stage 1 model), we modelled the presence/absence of each species across the FIA plot locations. In the second stage (Stage 2 model), we modelled the basal area per hectare of each species conditional on its presence at FIA plot locations. All models were fit using Markov chain Monte Carlo (MCMC) simulation using the 'spOccupancy' (Doser et al., 2022; Stage 1) or 'spAbundance' (Doser et al., 2024; Stage 2) packages. More detail on our modelling approach can be found in [Appendix S1](#).

2.4 | Candidate models

Eight candidate models were considered for both the Stage 1 (presence/absence) and Stage 2 (abundance) models varying both the environmental variables considered and the random effect structure ([Appendix S2, Table S1](#)). Six combinations of our eight environmental variables were considered including three additive combinations of climate, climate + physiographic, and climate + physiographic + biophysical variables, and another three sets incorporating quadratic terms for all variables considered. Stage 2 models also included measures of BAH and QMD as a measure of forest density and size.

In addition to alternative environmental variable sets, we considered three random effect structures including fixed effects only, a non-spatial random effect and a spatial random effect ([Appendix S2, Table S1](#)). No random effects were included under the fixed effects only model (the u_{ij} and v_{ij} terms are dropped from Equations 2 and 4 respectively in [Appendix S1](#)) equivalent to modelling all species independently (except for the hierarchical approach to estimate

regression coefficients as in a multi-species occupancy or abundance model). The non-spatial random effect accounts for residual covariation among species but assumes that the random effects are not spatially structured (i.e. that there is no residual spatial autocorrelation). Finally, the spatial random effect model accounts for spatially correlated residual covariation among species.

2.5 | Model selection

Bayesian model selection was applied to identify the best predicting Stage 1 and 2 models based on threefold cross-validation applying the log posterior predictive density scoring rule (log score) to evaluate out-of-sample prediction (Hooten & Hobbs, 2015). Our interest was in the ability of the candidate models to predict the richness (Stage 1) or total basal area per hectare (Stage 2) of the subset 11 species historically associated with American chestnut.

We applied a modified version of the community log score (Itter et al., 2024) as our model performance metric for both the Stage 1 and 2 models. We used the 'poibin' R package (Hong & R Core Team, 2024) to evaluate the posterior predictive density for out-of-sample observations of chestnut-associated species richness. We used the 'lognorm' R package (Wutzler, 2021) to approximate the posterior predictive density for out-of-sample observations of total basal area of chestnut-associated species. Additional details on the log score approximation for the Stage 1 and 2 models are provided in [Appendix S1](#).

We applied a two-phase model selection procedure for both the Stage 1 and 2 models. In the first phase, we evaluated the six potential variable sets applying a fixed effects only model structure ([Appendix S2, Table S1](#)). The goal of phase 1 was to identify the best predicting set of environmental variables before considering different random effect structures. In the second phase, we evaluated the fixed effects only, non-spatial random effects, and spatial random effects structures each using the environmental variable set identified under phase one. The goal of phase 2 was to identify the best predicting random effect structure conditional on the best predicting set of environmental variables.

The two-phase model selection procedure was implemented to avoid fitting an intractable number of computationally demanding Bayesian JSDMs. Note that with six potential variable sets and three random effect structures there are 18 possible model combinations for each stage (36 models total $\times$ 3 cross-validation folds). Further, flexible random effect structures, particularly those that incorporate spatial dependence, may be confounded with environmental variables affecting inference on the fixed effects in JSDMs (Van Ee et al., 2022). Thus, there is a statistical justification for separating the evaluation of fixed and random effects.

2.6 | Cross-validation

We developed a cross-validation scheme to mirror extrapolation in climate space from current to future conditions. To do so we

first fit an intercept only spatial factor JSJM to estimate the effective spatial range of potential spatial autocorrelation across the study plots using the 'sfJSJM' function from the R package 'spOccupancy' (Doser et al., 2022). We then created spatial blocks at the scale of the estimated effective spatial range (~155 km) over the study area using the 'cv_spatial' function from the 'blockCV' package (Valavi et al., 2019). Using the 'kmeans' function from the 'stats' package (R Core Team, 2024), we performed a Hartigan-Wong k means clustering analysis on the mean minimum temperature of each month of the growing season (May–September) across each block weighted by its overlap with the temperature rasters to assign blocks to three cross-validation folds. Fold assignment was further refined using the empirical cumulative distribution function of mean minimum growing season temperature of plots within each block until the number of points in each fold was roughly equal (Figure 1).

2.7 | Prediction

The best predicting Stage 1 and 2 candidate models, based on model selection procedure, were refit using the full set of FIA plot observations. We refer to these refit models as the final Stage 1 and 2 models respectively. The final Stage 1 model was used to predict the presence/absence of all study species under current and future climate scenarios. The final Stage 2 model was used to predict the total basal area of all study species conditional on their presence (Stage 1 model predictions) under current and future climate scenarios.

Predictions were made across a dense collection of point locations equivalent to the centroid of the TerraClimate grid (4 km resolution; Abatzoglou et al., 2018) within states overlapping the pre-blight range of American chestnut (hereafter 'study region'; Little, 1971). Predictions were made for a low-to-intermediate forest density and over different climate scenarios. Specifically, the BAH and QMD were set to values representative of the early development stages of eastern temperate forests based on a historic stocking guide for oak-hickory forests in the eastern US and close to the average for these values across plots (BAH = 19.51 m²/ha; QMD = 20.32 cm; Gingrich, 1971). Climate variables were set to TerraClimate estimates under current climatic conditions and under climate conditions corresponding to +2°C and +4°C global warming above preindustrial levels (Qin et al., 2020), independent of any specific year or emissions scenario. The combined BAH of the 11 chestnut-associated species was divided by the total BAH of all 39 species to get the proportion of chestnut-associated species abundance at each prediction location.

2.7.1 | Suitable sites for conservation plantings

To identify feasible sites for American chestnut plantings across our study region, we calculated the net change in the proportion of chestnut-associated species abundance for each warming scenario.

We applied a filter to only include locations with >0 net change such that the site is likely to be more suitable for American chestnut-associated species with climate change. Additionally, we classified each prediction location as either increasing or decreasing in proportional predicted chestnut-associated species abundance across the study region and within the historic range of American chestnut from current to +2°C and +4°C warming scenarios.

To assess the predictions made by our model within active conservation lands, we used the USGS Protected Areas Database (USPAD; USGS 2024). We filtered this dataset to only include areas designated as statuses '1' or '2' managed for biodiversity or mostly for biodiversity respectively within the states making up our study region. For each conservation area, we calculated the median proportion of chestnut-associated species abundance from prediction locations overlapping it under current, +2°C and +4°C warming scenarios. Conservation areas which did not have overlapping prediction locations used a single nearest location. We then calculated the net change in proportional chestnut-associated species abundance from current to +2°C and +4°C warming scenarios and identified areas predicted to increase as promising for conservation plantings.

Using predicted localities which overlapped these protected areas, we highlighted two examples of conservation areas which could be viable locations for restoration and assisted migration plantings. Both are predicted to have an increase in proportional chestnut-associated species abundance from current to +2°C and +4°C climate scenarios, with one falling outside and one within the historic range of American chestnut.

3 | RESULTS

3.1 | FIA dataset

The final dataset contained 25,416 FIA plots with abundance data for 39 common eastern temperate tree species (see Appendix S2 for complete list) spanning the pre-blight range of American chestnut which were included in modelling (Figure 1b). Analysis of the literature, supported by the fossil pollen record at the genus level (Appendix S2, Table S2), revealed that 11 of these species are commonly associated with American chestnut: *Acer rubrum*, *Betula lenta*, *Carya cordiformis*, *Carya glabra*, *Carya ovata*, *Fagus grandifolia*, *Quercus alba*, *Quercus coccinea*, *Quercus prinus*, *Quercus rubra*, and *Quercus velutina*. The dataset encompassed a broad climate dataset; minimum growing season temperatures ranged from 2.8°C to 16.9°C (mean = 9.52°C) and total growing season precipitation ranged from 385.7 mm to 827.9 mm (mean = 525.4 mm).

3.2 | Model selection

The combination of covariates providing the best prediction among the six fixed effects only Stage 1 models included linear and quadratic terms for climate and physiographic variables. Stage

2 fixed effects only models that included quadratic terms (three candidate models) all had similar predictive performance across the three cross-validation folds (log scores were within 0.8% of each other). As such, we applied the most parsimonious of the three candidate covariate sets including linear and quadratic terms for climate variables.

For both the Stage 1 and 2 random effects models, candidate models including the best predicting set of environmental covariates from the fixed effects models and non-spatial random effects yielded the best predictive performance compared to fixed effects only and spatial random effect candidate models. Subsequently, the non-spatial random effect candidate models were used to predict the presence/absence (Stage 1) and abundance conditional on presence (Stage 2) of species historically associated with chestnut under current and future climate scenarios (Appendix S2, Table S1). The most important covariates across our 11 chestnut-associated species in the final Stage 1 model included, organic soil content, pH, slope, and minimum growing season temperature and the quadratic of the minimum growing season temperature (Appendix S2, Figure S1). In the final Stage 2 model the most important variables included total BAH and QMD of all 39 species modelled (Appendix S2, Figure S2).

3.3 | Suitable sites for conservation plantings

Predictions of a high proportion of chestnut-associated species' abundance across the eastern U.S. under current conditions correspond well with the historic range of American chestnut, with highest abundance around the Appalachians. One difference between the currently observed and predicted range is an overprediction in northern New England and Michigan. Another is an underprediction across most of the southern historic range of American

chestnut (Figure 2a). Under +2°C (Figure 2b) and +4°C (Figure 2c) warming scenarios, applied models predicted a progressive poleward shift in proportional chestnut-associated species abundance into the northeast, with moderate to high predictions in the south limited to higher elevation regions of the Appalachians. Under both future warming scenarios, a net loss can be seen in the mid-Atlantic and south, while gains are limited to the Northeast and northern Michigan (Figure 3).

Under a +2°C warming scenario, proportional chestnut-associated species abundance is predicted to decrease across 78%, increase in 19%, and remain stable at zero proportional abundance in 3% of sites within the study region. When the analysis is limited to the historic range of American chestnut, there is a decrease in 92% and an increase in only 8% of predicted sites. The results are similar under a +4°C scenario for both the full region (84% decrease, 13% increase, 3% stable) and the historic range of American chestnut (98% decrease and 1% increase). The median net decrease in proportional chestnut-associated species abundance across prediction locations within its historic range was only 9.8% under a +2°C scenario and 24% under a +4°C scenario, suggesting the pace of species turnover at these sites may not be so rapid as to prevent within-range conservation plantings of American chestnut.

Locations with a net increase in predicted proportional chestnut-associated species abundance relative to current conditions identified the most promising locations for experimental conservation plantings across the study area as climate change progresses (Figure 3c, Figure 3d). Under the +2°C warming scenario this overlaps 3,765,498 hectares from 5598 unique conservation areas. Under a +4°C warming scenario this drops to 3,260,084 hectares representing 2988 unique conservation areas (Appendix S2, Figure S3). The names and administrative information of each conservation area are listed in Appendix S3.

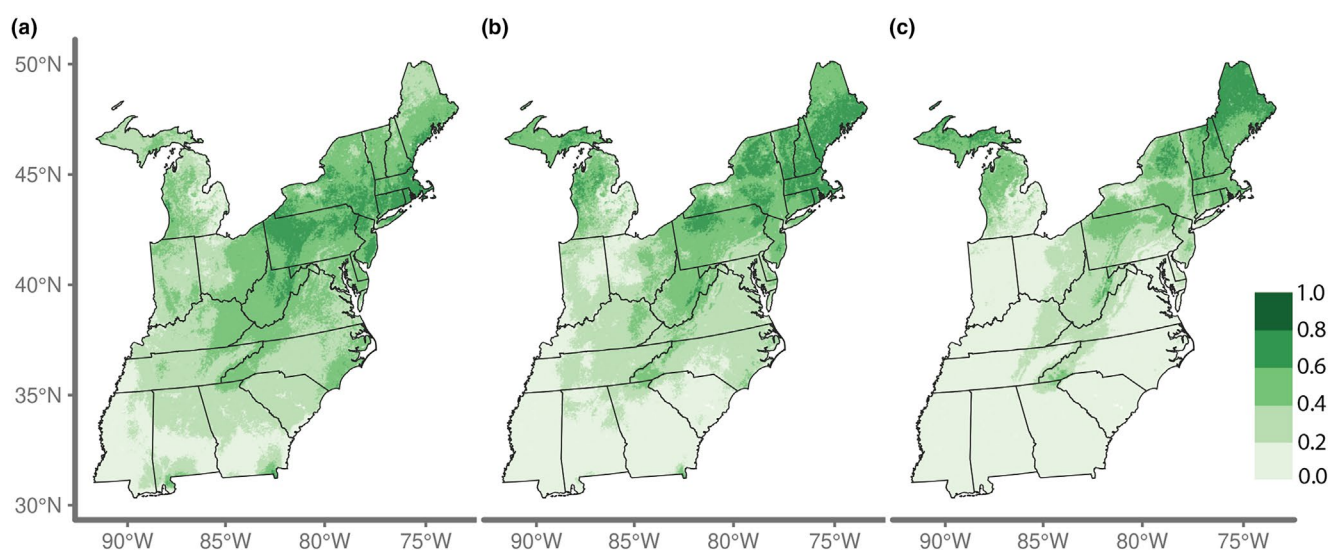
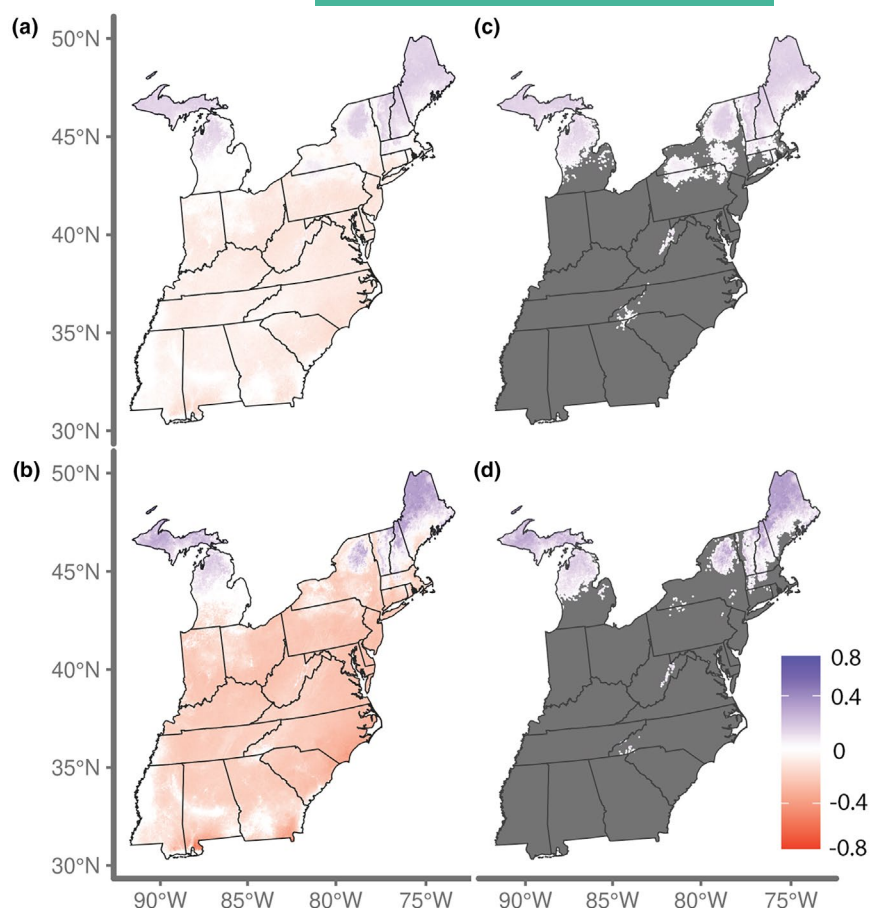


FIGURE 2 Proportion of basal area/ha of 11 tree species associated with American chestnut out of total basal area/ha of 39 modelled species under (a) current conditions, (b) +2°C warming, and (c) +4°C warming. Areas of predicted high abundance for chestnut-associated species are dark green, while areas of low predicted abundance are light green.

FIGURE 3 Predicted change in the proportion of basal area per hectare (BAH) of 11 species associated with American chestnut out of total BAH of all 39 modelled species under (a) +2°C warming and (b) +4°C warming relative to current conditions. Areas in blue represent a net gain in chestnut-associated species at a location, while red is a net decrease. Most promising areas for climate resilient American chestnut plantings predicted to have a net increase (>0) difference in chestnut-associated species abundance under (c) +2°C warming and (d) +4°C warming.



3.4 | Highlighted examples

The first example of suitable locations for restoration and assisted migration plantings within existing conservation lands focuses on Baxter State Park in northern Maine, an 84,782 hectare area managed by the Baxter State Park Authority. American chestnut is not known to occur here, but three of its associates, *A. rubrum*, *F. grandifolia* and *Q. rubra* do (Mittelhauser et al., 2016). Our model predicts increasing suitability for these and other chestnut-associated species under both future climate scenarios at prediction locations overlapping the park (Figure 4a).

The second example is Cheat Wildlife Management Area, a 32,026 hectare area managed by the West Virginia Division of Natural Resources within Monongahela National Forest. This area falls within the historic range of American chestnut, where it is known to occur along with most of its associates (Thomas-Van Gundy & Strager, 2011). Our model found that proportional chestnut-associated species abundance marginally increases in the area, suggesting stability for American chestnut and its associates under both future climate scenarios (Figure 4b).

4 | DISCUSSION

Since the blight began decimating populations of American chestnut, significant effort has been made towards its restoration across its

historical range. More recently, this work has been informed by models which predict potential habitat for the species under current and future climate conditions both across its entire historical range (e.g. Adeyemo & Granger, 2023; Barnes & Delborne, 2019; Noah et al., 2021), and at smaller scales (Henderson et al., 2022, 2023; Klaus & Klaus, 2025; Tulowiecki, 2020). However, these existing models rely on incomplete post-blight distributional data, lack true absences, include only abiotic predictor variables, and focus solely on presence/absence rather than potential abundance of American chestnut. The two-stage JSDM approach we used in this study addresses these pitfalls and importantly integrates statistical dependence among species in a forest (Clark et al., 2014; Pollock et al., 2014; Warton et al., 2015). The abundance of 11 species historically associated with American chestnut can act as a proxy for its distribution across the landscape providing a clearer picture of where this species may thrive today and under future climate conditions.

Our approach to modelling a functionally extinct species has several novel components. First, this is the first study which takes advantage of the joint probability of chestnut-associated species to estimate the community of a species which has been drastically reduced in occurrence and abundance across the landscape. Second, our model selection process was tailored to match the predictive task we set out to accomplish. This involved the use of a novel cross-validation strategy splitting blocks into folds which mirror the

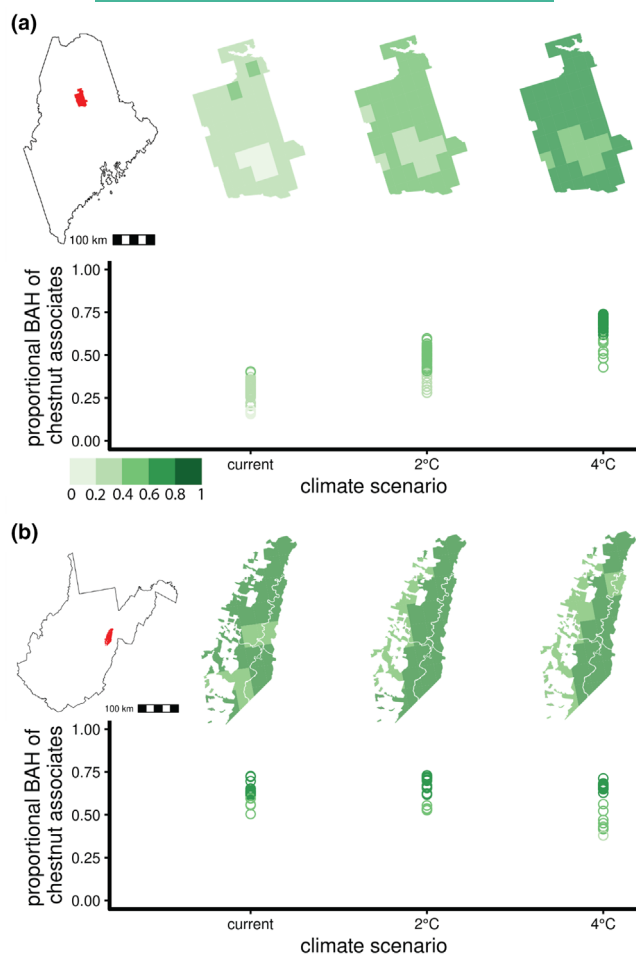


FIGURE 4 Example conservation areas for American chestnut showing an (a) increase (Baxter State Park, Maine) and (b) relative stability (Cheat Wildlife Management Area, West Virginia) in the proportion of 11 species associated with American chestnut out of total BAH of all 39 modelled species. Each circle on the graph represents the predicted value of a point within the 4km resolution grid overlapping the area's footprint mapped above for each climate scenario.

extrapolation in climate space being made to +2°C and +4°C warming scenarios. Through this approach, log scores identified the non-spatial random effect models as the best performing in both Stage 1 and Stage 2. While spatial random effects generally improve (greater accuracy, less uncertainty) JSDM predictions (Doser et al., 2023), previous studies have found relatively short effective spatial ranges associated with FIA plot data (May & Finley, 2025). The applied cross-validation approach was designed to control for the potential of spatial autocorrelation to bias model predictions when predicting across large geographic areas (Ploton et al., 2020) by creating folds beyond the effective spatial range. As such, it is less surprising to find that the non-spatial random effect models outperform the spatial random effect versions. This points to the importance of constructing a cross-validation approach that follows from the predictive objective of the analysis (Roberts et al., 2017).

One potential limitation to our approach is that species assemblages of North American forests are not fixed (Williams et al., 2001)

and may respond differently to a changing climate (Gougherty et al., 2024; Hobbs et al., 2018). This introduces uncertainty about the strength of the associations between the 11 species we selected as associates for American chestnut across space and time. However, because co-occurrence patterns often arise from shared environmental filtering and underlying trait similarity, these assemblages can still function as useful indirect indicators of suitable habitat conditions (Ovaskainen et al., 2017). Moreover, our identification approach grounded in historical literature and supported by genus-level co-occurrence in the fossil record likely captures recent ecological relationships, providing a robust approximation of suitability. Because American chestnut is functionally extinct and has been extirpated from much of its historical range, direct observations of its habitat preferences are limited; therefore, our approach using associated species provides one of the only feasible ways to infer its potential suitability.

Results from our model identified areas of highest proportional chestnut-associated species abundance in the Appalachian region but predicted relatively low associate species abundance in the southern part of American chestnut's historical range. This result is validated by limited American chestnut understory trees and sprouts in that part of the pre-blight range from FIA data (Dalgleish et al., 2016) and reinforced by similar conclusions in other distribution models (Barnes & Delborne, 2019; Noah et al., 2021). In contrast, studies modelling chestnut occurrence in the southern portion of the historical range specifically have found suitable habitat within the region (Fei et al., 2007; Laport et al., 2022). The southern range edge of American chestnut was reportedly retracting northwards since at least the 19th century, likely due to an early introduction of a separate pathogen, *Phytophthora cinnamomi* (Russell, 1987). This retraction may have led to under sampling of chestnut that was once more prevalent in the region and only captured in models more closely investigating occurrence in the southern portion of the range. Furthermore, American chestnut sampled from the southern portion of its historical range is known to be more genetically diverse and distinct from northeast and central populations (Sandercock et al., 2022; Spriggs & Fertakos, 2021). Therefore, southern populations of American chestnut may be more locally adapted to unique abiotic conditions not captured by our model or have weaker associations with the associated species used in our study, leading to our underprediction. Although we did not predict high associate species abundance in the lower portion of its range, remnant American chestnut occurring here contains high genetic and adaptive diversity that requires preservation (Sandercock et al., 2024).

Our model also predicted relatively high chestnut-associated species abundance in the Northeast and northern Michigan outside of American chestnut's pre-blight range. The northern edge of American chestnut and many other eastern temperate tree species is mainly limited by its sensitivity to frost (Gurney et al., 2011; Russell, 1987; Saielli et al., 2012), and as northern latitudes experience increases in temperature due to climate change, many northern tree species have been documented shifting northward (Woodall et al., 2009), which could explain this result. At the time of the blight's arrival, American

chestnut was continuing its post-glacial spread north into Michigan, and there have been many successful translocated plantings of the tree in the western portion of the lower peninsula (Brewer, 1995). Chestnuts planted in western Maine north of the natural range of the species have also shown success in establishing, reproducing, and spreading (Mazurowski et al., 2022). But challenges remain towards widespread implementation. Assisted migration plantings at the northern range edge will require plant material adapted to the conditions there (Pike et al., 2020). But populations of American chestnut, especially reproductive individuals, are lower along its northern range edge already due to the blight (Van Drunen et al., 2017), meaning limited material is available in germplasm storage. To support successful restoration of American chestnut at scale at and above its northern range limit, production of plant material of appropriate provenance will need to be significantly increased.

We identified a poleward shift in proportional American chestnut-associated species abundance and loss in the mid-Atlantic and southern portions of its historical range under +2°C and +4°C warming scenarios. Losses in the mid-Atlantic and southern portions of chestnut's historical range are substantial, with projected decreases in proportional chestnut-associated species abundance by 92% under +2°C and 98.4% under +4°C warming scenarios. This result aligns with that found by other models of American chestnut at this scale (Barnes & Delborne, 2019; Noah et al., 2021) and supports a restoration strategy that acknowledges northward shifts in tree populations (Clark et al., 2022). Eastern temperate forests have faced similar warming in the recent past, with global mean temperatures increasing by ~4°C from the last glacial maximum ~21,000 years ago to the start of the present Holocene interglacial period ~12,000 years ago (Clark et al., 2016). Although eastern temperate tree species have historically shifted their ranges in response to changes in climate, current change is an order of magnitude faster (Clark et al., 2016), potentially exceeding the rate at which tree species can move, especially slow dispersers like American chestnut (Davis, 1983). Increasing drought, more frequent wildfires, the spread of invasive species, and other climate change driven disturbances to eastern temperate forests that we were not able to account for in our model add additional concern for the conservation of this species.

The loss of suitable habitat within chestnut's historical range, its slow migration rate relative to changing climate conditions, and the limited number of seed-producing mature trees due to the blight mean assisted migration plantings will be required to ensure the long-term survival of the species. This type of adaptive management has long been recommended for forest management in the context of climate change (Millar et al., 2007). In the last few decades, many achievements have been made towards the production of blight-resistant hybrid (Jacobs, 2007) and more recently transgenic (Newhouse & Powell, 2021) chestnuts for use in experimental plantings. Assisted migration plantings of blight-resistant chestnuts have already shown success in some of the areas predicted to have high future suitability, and land designated for the conservation of biodiversity overlaps these areas that can be used for future experiments

by land managers. For example, American chestnut planted in northern New Hampshire performed well with a high survival-growth response compared to other eastern temperate tree species considered for assisted migration (Clark et al., 2022).

Our predictions are made at a broad scale and are therefore missing relevant local scale predictors, but investigation at the scale of conservation areas gives a general idea on how the forest community may respond to future climate change. In Baxter State Park in northern Maine, we found the proportional chestnut-associated species abundance increased from around 30% to 70% under a +4°C warming scenario. This park will likely be a good candidate for potential experimental assisted migration plantings of American chestnut as climate change progresses due to its predicted suitability for the species historically associated with it.

While we predict a loss in the proportional abundance of species associated with American chestnut throughout most predicted locations within its historic range, this does not warrant abandoning restoration plantings and management in these areas altogether. The median net decrease was only 9.8% under a +2°C scenario and 24% under a +4°C scenario, suggesting a more gradual turnover in the composition of these forests and their likely persistence by the turn of the century. For example, within Cheat Wildlife Management Area in West Virginia, the proportional chestnut-associated species abundance remains somewhat stable at around 60% under both climate scenarios. As this area falls within the historical range of American chestnut and predicted suitability for its associated species, it will likely serve as a refugia for the species as our climate continues to change and should be a focus for restoration plantings. Additional promising locations for assisted migration plantings include areas already designated for conservation within upstate New York, Michigan, Vermont and New Hampshire.

A limitation of our study is that habitat suitability was inferred from chestnut-associated species, which may not fully capture the future distribution of American chestnut, especially at the broad scale of our predictions compared to local conservation planning. Additionally, our model was restricted to the contiguous United States because FIA plot data do not extend into Canada. Consequently, we were unable to evaluate habitat suitability in regions of southern Ontario, Quebec, and the Canadian Maritimes, where forest communities capable of supporting assisted migration plantings of American chestnut likely occur and may become increasingly important under future climate conditions. Future research should therefore assess the current and projected suitability of American chestnut across southern Canada to better inform conservation and restoration efforts.

As climate change continues to reshape eastern temperate forests, predictive modelling will become increasingly important for guiding adaptive restoration strategies. For American chestnut, our results suggest that restoration efforts should prioritize both conservation plantings within climatically stable portions of the historical range and assisted migration plantings in northern areas projected to become more suitable under future warming. By identifying nearly three thousand protected areas that are expected to

maintain or increase suitability for chestnut-associated forest communities under a +4°C scenario, our study provides land managers with a spatially explicit framework for prioritizing field trials, seed deployment, and long-term restoration investments. These locations offer opportunities to test blight-resistant trees, evaluate provenance performance under novel climates, and establish restoration plantings in landscapes already managed for biodiversity conservation. Together, our results provide practical guidance for increasing the likelihood that American chestnut restoration efforts remain successful under ongoing climate change.

AUTHOR CONTRIBUTIONS

Matthew E. Fertakos, Bethany A. Bradley, and Malcolm S. Itter conceived the ideas and designed methodology; Matthew E. Fertakos, Malcolm S. Itter, and Jefferey W. Doser developed the model structure; Matthew E. Fertakos and Malcolm S. Itter performed the analyses; Matthew E. Fertakos led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGEMENTS

We thank the staff at the Unity high-performance computing cluster for assistance with model implementation. This work was supported by an NSF GRFP No. 1938059 awarded to M. Fertakos and by the US Geological Survey Northeast Climate Adaptation Science Center (NE CASC) award number G23AC00519.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data and code available from Zenodo <https://doi.org/10.5281/zenodo.21419542> (Fertakos, 2026).

ORCID

Matthew E. Fertakos  <https://orcid.org/0000-0002-8121-0039>

Bethany A. Bradley  <https://orcid.org/0000-0003-4912-4971>

Jeffery W. Doser  <https://orcid.org/0000-0002-8950-9895>

Malcolm S. Itter  <https://orcid.org/0000-0002-0184-2731>

REFERENCES

- Abatzoglou, J. T., Dobrowski, S. Z., Parks, S. A., & Hegewisch, K. C. (2018). TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. *Scientific Data*, 5, 170191. <https://doi.org/10.1038/sdata.2017.191>
- Adeyemo, S. M., & Granger, J. J. (2023). Habitat suitability model and range shift analysis for American Chestnut (*Castanea dentata*) in the United States. *Trees, Forests and People*, 11, 100360. <https://doi.org/10.1016/j.tfp.2022.100360>
- Anagnostakis, S. L. (1995). The pathogens and pests of chestnuts. In J. A. Callow, J. H. Andrews, & I. C. Tommerup (Eds.), *Advances in botanical research* (Vol. 21, pp. 125–145). Academic Press. [https://doi.org/10.1016/S0065-2296\(08\)60011-7](https://doi.org/10.1016/S0065-2296(08)60011-7)
- Barnes, J. C., & Delborne, J. A. (2019). Rethinking restoration targets for American chestnut using species distribution modeling. *Biodiversity and Conservation*, 28(12), 3199–3220. <https://doi.org/10.1007/s10531-019-01814-8>
- Braun, E. L. (1950). *Deciduous forests of eastern North America*. Blakiston Company.
- Brewer, L. G. (1995). Ecology of survival and recovery from blight in American chestnut trees (*Castanea dentata* (marsh.) Borkh.) in Michigan. *Bulletin of the Torrey Botanical Club*, 122(1), 40–57. <https://doi.org/10.2307/2996402>
- Burke, K. L. (2012). Niche contraction of American chestnut in response to chestnut blight. *Canadian Journal of Forest Research*, 42(3), 614–620. <https://doi.org/10.1139/x2012-002>
- Clark, J. S., Gelfand, A. E., Woodall, C. W., & Zhu, K. (2014). More than the sum of the parts: Forest climate response from joint species distribution models. *Ecological Applications*, 24(5), 990–999. <https://doi.org/10.1890/13-1015.1>
- Clark, P. U., Shakun, J. D., Marcott, S. A., Mix, A. C., Eby, M., Kulp, S., Levermann, A., Milne, G. A., Pfister, P. L., Santer, B. D., Schrag, D. P., Solomon, S., Stocker, T. F., Strauss, B. H., Weaver, A. J., Winkelmann, R., Archer, D., Bard, E., Goldner, A., ... Plattner, G.-K. (2016). Consequences of twenty-first-century policy for multi-millennial climate and sea-level change. *Nature Climate Change*, 6(4), 360–369. <https://doi.org/10.1038/nclimate2923>
- Clark, P. W., Freeman, A. J., D'Amato, A. W., Schaberg, P. G., Hawley, G. J., Evans, K. S., & Woodall, C. W. (2022). Restoring a keystone tree species for the future: American chestnut assisted migration plantings in an adaptive silviculture experiment. *Forest Ecology and Management*, 523, 120505. <https://doi.org/10.1016/j.foreco.2022.120505>
- Cogbill, C. V., Burk, J., & Motzkin, G. (2002). The forests of presettlement New England, USA: Spatial and compositional patterns based on town proprietor surveys. *Journal of Biogeography*, 29(10–11), 1279–1304. <https://doi.org/10.1046/j.1365-2699.2002.00757.x>
- Dalgleish, H. J., Nelson, C. D., Scrivani, J. A., & Jacobs, D. F. (2016). Consequences of shifts in abundance and distribution of American chestnut for restoration of a foundation forest tree. *Forests*, 7(1), Article 1. <https://doi.org/10.3390/f7010004>
- Dalgleish, H. J., & Swihart, R. K. (2012). American chestnut past and future: Implications of restoration for resource pulses and consumer populations of eastern U.S. Forests. *Restoration Ecology*, 20(4), 490–497. <https://doi.org/10.1111/j.1526-100X.2011.00795.x>
- Davis, D. E. (2005). Historical significance of American chestnut to Appalachian culture and ecology. In K. C. Steiner & J. E. Carlson (Eds.), *Proceedings of the conference on restoration of American chestnut to forest lands* (pp. 53–60). U.S. Department of the Interior, National Park Service.
- Davis, M. B. (1983). Quaternary history of deciduous forests of Eastern North America and Europe. *Annals of the Missouri Botanical Garden*, 70(3), 550–563. <https://doi.org/10.2307/2992086>
- Dominguez, S., & Goring, S. (2023). Working with the Neotoma paleoecology database. <https://github.com/NeotomaDB/neotoma2>
- Doser, J. W., Finley, A. O., & Banerjee, S. (2023). Joint species distribution models with imperfect detection for high-dimensional spatial data. *Ecology*, 104(9), e4137. <https://doi.org/10.1002/ecy.4137>
- Doser, J. W., Finley, A. O., Kéry, M., & Zipkin, E. F. (2022). spOccupancy: An R package for single-species, multi-species, and integrated spatial occupancy models. *Methods in Ecology and Evolution*, 13(8), 1670–1678. <https://doi.org/10.1111/2041-210X.13897>
- Doser, J. W., Finley, A. O., Kéry, M., & Zipkin, E. F. (2024). spAbundance: An R package for single-species and multi-species spatially explicit abundance models. *Methods in Ecology and Evolution*, 15(6), 1024–1033. <https://doi.org/10.1111/2041-210X.14332>
- Doser, J. W., Itter, M. S., Domke, G. M., & Finley, A. O. (2025). Multivariate spatial models for small area estimation of species-specific forest inventory parameters. *Forest Ecology and Management*, 597, 123112. <https://doi.org/10.1016/j.foreco.2025.123112>

- Ellison, A. M., Bank, M. S., Clinton, B. D., Colburn, E. A., Elliott, K., Ford, C. R., Foster, D. R., Kloeppel, B. D., Knoepp, J. D., Lovett, G. M., Mohan, J., Orwig, D. A., Rodenhouse, N. L., Sobczak, W. V., Stinson, K. A., Stone, J. K., Swan, C. M., Thompson, J., Von Holle, B., & Webster, J. R. (2005). Loss of foundation species: Consequences for the structure and dynamics of forested ecosystems. *Frontiers in Ecology and the Environment*, 3(9), 479–486. [https://doi.org/10.1890/1540-9295\(2005\)003\[0479:LOFSCF\]2.0.CO;2](https://doi.org/10.1890/1540-9295(2005)003[0479:LOFSCF]2.0.CO;2)
- Fei, S., Schibig, J., & Vance, M. (2007). Spatial habitat modeling of American chestnut at Mammoth Cave National Park. *Forest Ecology and Management*, 252(1), 201–207. <https://doi.org/10.1016/j.foreco.2007.06.036>
- Fertakos, M. E. (2026). Data and code from: “joint species distribution modeling for restoration of American Chestnut (*Castanea dentata*) in the United States under current and future climate conditions”. Zenodo. <https://doi.org/10.5281/zenodo.21419542>
- Gingrich, S. F. (1971). Stocking, growth, and yield of oak stands. In *Oak symposium proceedings. 1971 August 16-20* (pp. 65–73). U.S. Department of Agriculture, Forest Service, Northeastern Forest Experiment Station. <https://research.fs.usda.gov/treearch/14517>
- Gougherty, A. V., Prasad, A. M., Peters, M. P., Matthews, S. N., & Adams, B. T. (2024). Climate change and the emergence of no-analog forest assemblages in North America. *Global Change Biology*, 30(12), e17605. <https://doi.org/10.1111/gcb.17605>
- Gurney, K. M., Schaberg, P. G., Hawley, G. J., & Shane, J. B. (2011). Inadequate cold tolerance as a possible limitation to American chestnut restoration in the Northeastern United States. *Restoration Ecology*, 19(1), 55–63. <https://doi.org/10.1111/j.1526-100X.2009.00544.x>
- Hanberry, B. B., & Nowacki, G. J. (2016). Oaks were the historical foundation genus of the east-central United States. *Quaternary Science Reviews*, 145, 94–103. <https://doi.org/10.1016/j.quascirev.2016.05.037>
- Henderson, A. F., Santoro, J. A., & Kremer, P. (2022). Ensemble modeling for American chestnut distribution: Locating potential restoration sites in Pennsylvania. *Frontiers in Ecology and Evolution*, 10, 942766. <https://doi.org/10.3389/fevo.2022.942766>
- Henderson, A. F., Santoro, J. A., & Kremer, P. (2023). Impacts of spatial scale and resolution on species distribution models of American chestnut (*Castanea dentata*) in Pennsylvania, USA. *Forest Ecology and Management*, 529, 120741. <https://doi.org/10.1016/j.foreco.2022.120741>
- Hobbs, R. J., Valentine, L. E., Standish, R. J., & Jackson, S. T. (2018). Movers and stayers: Novel assemblages in changing environments. *Trends in Ecology & Evolution*, 33(2), 116–128. <https://doi.org/10.1016/j.tree.2017.11.001>
- Hong, Y., & R Core Team. (2024). poibin: The Poisson binomial distribution [R package]. <https://doi.org/10.32614/CRAN.package.poibin>
- Hooten, M. B., & Hobbs, N. T. (2015). A guide to Bayesian model selection for ecologists. *Ecological Monographs*, 85(1), 3–28. <https://doi.org/10.1890/14-0661.1>
- Itter, M. S., Kaarlejärvi, E., Laine, A.-L., Hamberg, L., Tonteri, T., & Vanhatalo, J. (2024). Bayesian joint species distribution model selection for community-level prediction. *Global Ecology and Biogeography*, 33(5), e13827. <https://doi.org/10.1111/geb.13827>
- Iverson, L. R., Prasad, A. M., Peters, M. P., & Matthews, S. N. (2019). Facilitating adaptive forest management under climate change: A spatially specific synthesis of 125 species for habitat changes and assisted migration over the eastern United States. *Forests*, 10(11), 989. <https://doi.org/10.3390/f10110989>
- Jacobs, D. F. (2007). Toward development of silvical strategies for forest restoration of American chestnut (*Castanea dentata*) using blight-resistant hybrids. *Biological Conservation, Forests in the Balance: Linking Tradition and Technology in Landscape Mosaics*, 137(4), 497–506. <https://doi.org/10.1016/j.biocon.2007.03.013>
- Klaus, J. M., & Klaus, N. A. (2025). Modelling the historic distribution and habitat of American chestnut (*Castanea dentata*) in Georgia, USA using edaphic and landform predictors. *Frontiers of Biogeography*, 18, 161937. <https://doi.org/10.21425/fob.18.161937>
- Korstian, C. F. (1927). *The natural replacement of blight-killed chestnut* (Clarence F., Stickel, P. W., & United States. Department of Agriculture (with U. S. Department of Agriculture, N. A. L.)). G.P.O. <http://archive.org/details/naturalreplaceme100kors>
- Laport, R. G., Brookover, Z. S., Christman, B. D., Ng, J., Philley, K., & Craddock, J. H. (2022). Environmental niche and demographic modeling of American chestnut near its southwestern range limit. *The American Midland Naturalist*, 188(2), 137–176.
- Little, E. L. (1971). *Digital representation of Atlas of United States trees*. U.S. Department of Agriculture. <https://pubs.usgs.gov/pp/p1650-a/>
- May, P. B., & Finley, A. O. (2025). Spatial-temporal prediction of forest attributes using latent Gaussian models and inventory data. *Spatial Statistics*, 69, 100917. <https://doi.org/10.1016/j.spasta.2025.100917>
- Mazurowski, J., Heinrich, B., Heinrich, L., Loeb, C., & Rives, R. (2022). The continued spread of a wild population of American chestnuts. *Northeastern Naturalist*, 29(3), 321–334. <https://doi.org/10.1656/045.029.0302>
- Millar, C. I., Stephenson, N. L., & Stephens, S. L. (2007). Climate change and forests of the future: Managing in the face of uncertainty. *Ecological Applications*, 17(8), 2145–2151. <https://doi.org/10.1890/06-1715.1>
- Mittelhauser, G. H., Bissell, J., Cameron, D., Dibble, A. C., Haines, A., Hoekwater, J., Lovit, M., & Megquier, A. (2016). *The plants of Baxter State Park* (1st ed.). University of Maine Press in association with Baxter State Park, Friends of Baxter State Park, and Maine Natural History Observatory.
- Newhouse, A. E., & Powell, W. A. (2021). Intentional introgression of a blight tolerance transgene to rescue the remnant population of American chestnut. *Conservation Science and Practice*, 3(4), e348. <https://doi.org/10.1111/csp2.348>
- Noah, P. H., Cagle, N. L., Westbrook, J. W., & Fitzsimmons, S. F. (2021). Identifying resilient restoration targets: Mapping and forecasting habitat suitability for *Castanea dentata* in Eastern USA under different climate-change scenarios. *Climate Change Ecology*, 2, 100037. <https://doi.org/10.1016/j.ecochg.2021.100037>
- Ovaskainen, O., Tikhonov, G., Norberg, A., Guillaume Blanchet, F., Duan, L., Dunson, D., Roslin, T., & Abrego, N. (2017). How to make more out of community data? A conceptual framework and its implementation as models and software. *Ecology Letters*, 20(5), 561–576. <https://doi.org/10.1111/ele.12757>
- Paillet, F. L. (2003). Chestnut: History and ecology of a transformed species. *Journal of Biogeography*, 29(10–11), 1517–1530. <https://doi.org/10.1046/j.1365-2699.2002.00767.x>
- Pike, C., Potter, K. M., Berrang, P., Crane, B., Baggs, J., Leites, L., & Luther, T. (2020). New seed-collection zones for the eastern United States: The eastern seed zone forum. *Journal of Forestry*, 118(4), 444–451. <https://doi.org/10.1093/jofore/fvaa013>
- Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., & Pélissier, R. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11(1), 4540. <https://doi.org/10.1038/s41467-020-18321-y>
- Poggiato, G., Münkemüller, T., Bystrova, D., Arbel, J., Clark, J. S., & Thuiller, W. (2021). On the interpretations of joint modeling in community ecology. *Trends in Ecology & Evolution*, 36(5), 391–401. <https://doi.org/10.1016/j.tree.2021.01.002>
- Pollock, L. J., Tingley, R., Morris, W. K., Golding, N., O'Hara, R. B., Parris, K. M., Vesk, P. A., & McCarthy, M. A. (2014). Understanding co-occurrence by modelling species simultaneously with a joint species distribution model (JSDM). *Methods in Ecology and Evolution*, 5(5), 397–406. <https://doi.org/10.1111/2041-210X.12180>

- Qin, Y., Abatzoglou, J. T., Siebert, S., Huning, L. S., AghaKouchak, A., Mankin, J. S., Hong, C., Tong, D., Davis, S. J., & Mueller, N. D. (2020). Agricultural risks from changing snowmelt. *Nature Climate Change*, 10(5), 459–465. <https://doi.org/10.1038/s41558-020-0746-8>
- R Core Team. (2024). *R: A language and environment for statistical computing* (Version 4.4.0) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ramcharan, A., Hengl, T., Nauman, T., Brungard, C., Waltman, S., Wills, S., & Thompson, J. (2018). Soil property and class maps of the conterminous United States at 100-meter spatial resolution. *Soil Science Society of America Journal*, 82(1), 186–201. <https://doi.org/10.2136/sssaj2017.04.0122>
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Aroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>
- Russell, E. W. B. (1987). Pre-blight distribution of *Castanea dentata* (Marsh.) Borkh. *Bulletin of the Torrey Botanical Club*, 114(2), 183–190. <https://doi.org/10.2307/2996129>
- Saielli, T. M., Schaberg, P. G., Hawley, G. J., Halman, J. M., & Gurney, K. M. (2012). Nut cold hardiness as a factor influencing the restoration of American chestnut in northern latitudes and high elevations. *Canadian Journal of Forest Research*, 42(5), 849–857. <https://doi.org/10.1139/x2012-033>
- Sandercock, A. M., Westbrook, J. W., Zhang, Q., & Holliday, J. A. (2024). A genome-guided strategy for climate resilience in American chestnut restoration populations. *Proceedings of the National Academy of Sciences of the United States of America*, 121(30), e2403505121. <https://doi.org/10.1073/pnas.2403505121>
- Sandercock, A. M., Westbrook, J. W., Zhang, Q., Johnson, H. A., Saielli, T. M., Scrivani, J. A., Fitzsimmons, S. F., Collins, K., Perkins, M. T., Craddock, J. H., Schmutz, J., Grimwood, J., & Holliday, J. A. (2022). Frozen in time: Rangewide genomic diversity, structure, and demographic history of relict American chestnut populations. *Molecular Ecology*, 31(18), 4640–4655. <https://doi.org/10.1111/mec.16629>
- Sittaro, F., Paquette, A., Messier, C., & Nock, C. A. (2017). Tree range expansion in eastern North America fails to keep pace with climate warming at northern range limits. *Global Change Biology*, 23(8), 3292–3301. <https://doi.org/10.1111/gcb.13622>
- Spriggs, E. L., & Fertakos, M. E. (2021). Evolution of *Castanea* in North America: Restriction-site-associated DNA sequencing and ecological modeling reveal a history of radiation, range shifts, and disease. *American Journal of Botany*, 108(9), 1692–1704. <https://doi.org/10.1002/ajb2.1726>
- Stanke, H., Finley, A. O., Weed, A. S., Walters, B. F., & Domke, G. M. (2020). rFIA: An R package for estimation of forest attributes with the US Forest Inventory and Analysis database. *Environmental Modelling & Software*, 127, 104664. <https://doi.org/10.1016/j.envsoft.2020.104664>
- Steiner, K. C., & Carlson, J. E. (2006). *Restoration of American Chestnut to Forest Lands: Proceedings of a conference and workshop held May 4–6, 2004 at the North Carolina arboretum*. U.S. Department of the Interior, National Park Service, National Capital Region, Center for Urban Ecology.
- Tälle, M., Ranius, T., & Öckinger, E. (2023). The usefulness of surrogates in biodiversity conservation: A synthesis. *Biological Conservation*, 288, 110384. <https://doi.org/10.1016/j.biocon.2023.110384>
- Terrain Tiles. (2026). Terrain Tiles dataset. Registry of Open Data on AWS. <https://registry.opendata.aws/terrain-tiles>
- Thomas-Van Gundy, M., & Strager, M. (2011). Site characteristics of American chestnut, oak, and hickory witness trees on the Monongahela National Forest, West Virginia. In S. Fei, J. M. Lhotka, J. W. Stringer, K. W. Gottschalk, & G. W. Miller (Eds.), *Proceedings of the 17th Central Hardwood Forest Conference* (Gen. Tech. Rep. NRS-P-78, pp. 208–218). U.S. Department of Agriculture, Forest Service, Northern Research Station.
- Tulowiecki, S. J. (2020). Modeling the historical distribution of American chestnut (*Castanea dentata*) for potential restoration in western New York State, US. *Forest Ecology and Management*, 462, 118003. <https://doi.org/10.1016/j.foreco.2020.118003>
- U.S. Geological Survey (USGS) Gap Analysis Project (GAP). (2024). Protected Areas Database of the United States (PAD-US) 4.1: U.S. Geological Survey data release. <https://doi.org/10.5066/P96WBCHS>
- Valavi, R., Elith, J., Lahoz-Monfort, J. J., & Guillera-Aroita, G. (2019). blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10, 225–232. <https://doi.org/10.1111/2041-210X.13107>
- Van Drunen, S. G., Schutten, K., Bowen, C., Boland, G. J., & Husband, B. C. (2017). Population dynamics and the influence of blight on American chestnut at its northern range limit: Lessons for conservation. *Forest Ecology and Management*, 400, 375–383. <https://doi.org/10.1016/j.foreco.2017.06.015>
- Van Ee, J. J., Ivan, J. S., & Hooten, M. B. (2022). Community confounding in joint species distribution models. *Scientific Reports*, 12(1), 12235. <https://doi.org/10.1038/s41598-022-15694-6>
- Warton, D. I., Blanchet, F. G., O'Hara, R. B., Ovaskainen, O., Taskinen, S., Walker, S. C., & Hui, F. K. C. (2015). So many variables: Joint modeling in community ecology. *Trends in Ecology & Evolution*, 30(12), 766–779. <https://doi.org/10.1016/j.tree.2015.09.007>
- Whitney, G. G. (1994). *From coastal wilderness to fruited plain: A history of environmental change in temperate North America, 1500 to the present*. Cambridge University Press.
- Williams, J. W., Grimm, E. C., Blois, J. L., Charles, D. F., Davis, E. B., Goring, S. J., Graham, R. W., Smith, A. J., Anderson, M., Arroyo-Cabales, J., Ashworth, A. C., Betancourt, J. L., Bills, B. W., Booth, R. K., Buckland, P. I., Curry, B. B., Giesecke, T., Jackson, S. T., Latorre, C., ... Takahara, H. (2018). The Neotoma paleoecology database, a multiproxy, international, community-curated data resource. *Quaternary Research*, 89(1), 156–177. <https://doi.org/10.1017/qua.2017.105>
- Williams, J. W., Shuman, B. N., & Webb, T., III. (2001). Dissimilarity analyses of late-quaternary vegetation and climate in Eastern North America. *Ecology*, 82(12), 3346–3362. [https://doi.org/10.1890/0012-9658\(2001\)082\[3346:DAOLQV\]2.0.CO;2](https://doi.org/10.1890/0012-9658(2001)082[3346:DAOLQV]2.0.CO;2)
- Wis, M. S., & Guisan, A. (2009). Do pseudo-absence selection strategies influence species distribution models and their predictions? An information-theoretic approach based on simulated data. *BMC Ecology*, 9(1), 8. <https://doi.org/10.1186/1472-6785-9-8>
- Woodall, C. W., Oswalt, C. M., Westfall, J. A., Perry, C. H., Nelson, M. D., & Finley, A. O. (2009). An indicator of tree migration in forests of the eastern United States. *Forest Ecology and Management*, 257(5), 1434–1444. <https://doi.org/10.1016/j.foreco.2008.12.013>
- Wutzler, T. (2021). lognorm: Functions for the lognormal distribution [R package]. <https://doi.org/10.32614/CRAN.package.lognorm>
- Zhang, C., Chen, Y., Xu, B., Xue, Y., & Ren, Y. (2020). Improving prediction of rare species' distribution from community data. *Scientific Reports*, 10(1), 12230. <https://doi.org/10.1038/s41598-020-69157-x>
- Zhu, K., Woodall, C. W., & Clark, J. S. (2012). Failure to migrate: Lack of tree range expansion in response to climate change. *Global Change Biology*, 18(3), 1042–1052. <https://doi.org/10.1111/j.1365-2486.2011.02571.x>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Supplemental Methods.

Appendix S2. Additional Tables and Figures.

Figure S1. Posterior beta estimates with 95% credible interval for each covariate included in the Stage 1 model separated by the 11 associated species. Species in which the credible interval crosses 0 (dotted line) are represented by black circles.

Figure S2. Posterior beta estimates with 95% credible interval for each covariate included in the Stage 2 model separated by the 11 associated species. Species in which the credible interval crosses 0 (dotted line) are represented by black circles.

Figure S3. Conservation lands predicted to have an increase in median proportional abundance of American chestnut-associated species under +2°C, or both +2°C and +4°C future climate conditions.

Table S1. Summary of the covariates and random effects included in each model and its corresponding log score used in model selection for both Stage 1 and Stage 2 models.

Table S2. Proportion of sites containing *Castanea* pollen in the Eastern Temperate Forest region across the last 22,000 years each genera co-occurs in. Genera of potential associated species are displayed. All proportions are high except for *Liriodendron*.

Appendix S3. 5598 unique conservation areas (by Unit_Nm) from the USGS Protected Areas Database (US-PAD) suitable for

restoration and experimental plantings of American chestnut shown in [Appendix S2](#), [Figure S3](#). Most suitable are those areas predicted to have an increase in the proportion of 11 species associated with American chestnut under both the +2C and +4C future climate scenarios ('yes' in columns clim_2C and clim_4C). Data descriptions for all column except 'clim_2C' and 'clim_4C' come from the US-PAD database: U.S. Geological Survey (USGS) Gap Analysis Project (GAP). (2024). Protected Areas Database of the United States (PAD-US) 4.1: U.S. Geological Survey data release. <https://doi.org/10.5066/P96WBCHS>.

How to cite this article: Fertakos, M. E., Bradley, B. A., Doser, J. W., & Itter, M. S. (2026). Joint species distribution modelling for restoration of American chestnut (*Castanea dentata*) in the United States under current and future climate conditions. *Journal of Applied Ecology*, 63, e70600. <https://doi.org/10.1111/1365-2664.70600>